



ARM Cortex-A9 processor (Courtesy: www.arm.com)

bands to gold and titanium jewelled versions with studded diamonds *et al!*

What makes wearable tech different—and difficult?

Wearable technology is special—not just because of the trendy feel but also because of the efforts that are required to create it. When designing and developing a wearable product, one has to remember that it has to be worn by a human comfortably. It should neither be too bulky nor too hot. Similarly, it should be safe. These basic requirements greatly influence the design goals of wearables.

Wearable technology requires very low-energy, battery-powered components to minimise heat dissipation and provide extended battery life, as it can be frustrating to change batteries or charge them frequently. Heat dissipation can lead to not just poor device performance but also discomfort for the user.

Size is another factor to be considered as the device is to be worn. For instance, the chip required to enable augmented reality in sunglasses will be a lot smaller than the chip in a mobile phone. Miniaturisation is becoming increasingly feasible, thanks to micro-electromechanical systems (MEMS) and the development of cost-efficient batch fabrication techniques for their manufacture. MEMS is making it possible to integrate components such as microprocessors, sensors and radio communication circuits into a single

integrated circuit (IC) or system-on-chip (SoC) implementation.

“Ensuring that the chips provide optimal performance whilst being compact and power-efficient is the key challenge that designers face. It is not just the usual electronic design challenges that one needs to work with. We want ‘always-on’ communications, apps and devices but there is still a lot of work to be done in areas such as interoperability and sustain-

ability,” says Guru Ganesan, managing director-India operations, ARM.

Low-power, low-footprint processors

Today, ARM’s Cortex series is popular for wearable applications. Nike Fuelband uses a Cortex-M3, the Fitbit family products are all ARM Cortex-M3 based, and advanced watch platforms like MyBasis and I’m Watch are mostly ARM based.

“We have a variety of products that cater to wearable products: Cortex-A application processor series, Cortex-R real-time processor series and Cortex-M processor series that caters to embedded applications. For most applications, Cortex-M series processors will be the most suitable, due to their very low power consumption and small footprint. Cortex-M0+ processor, for example, consumes as little as three microwatts per megahertz while occupying an area of just 0.01 mm². Despite their very small size and low power consumption, these processors incorporate sufficient intelligence to handle most wearable applications easily. Depending on the exact application, a Cortex-R series processor may also be incorporated to handle data communication tasks. Besides these, our Mali GPUs cater to the augmented reality space,” says Ganesan.

Intel too is working on creating processors for wearables. These will probably be half the size of the Atom processors used commonly in mobile

devices, and consume power in the range of a few microwatts.

Syncing with the wearer

Wearable devices use a variety of sensors ranging from gyroscopes and accelerometers to location sensors. Sensors in a health monitor might constantly measure vital body signs like pulse and temperature, while those in a fitness tool might monitor how many jumps you have done, how long you have run or how fast you are moving. Those in an augmented reality application might keep a close watch on how you move, bend or turn your head, to ensure that the display is in sync with the environment.

Sensors used in wearable technology can be broadly classified into body sensors and ambient sensors. MEMS technology has led to the development of miniature sensors that can be worn without much disturbance. Wearable sensors are sometimes combined with ambient sensors, which are embedded in the environment. A combination of body and ambient sensors could, for example, be used to unobtrusively monitor senior citizens in their home environment, collecting information on their activities, sleep, restroom visits, etc. The network, in turn, can convey alerts to remote caregivers if there is a variation in the usual behaviour pattern.

In early days, sensors were integrated using wires that ran through the garment. Today’s wearables, however, use wireless communications technologies like radio frequency (RF), near-field communications (NFC), Bluetooth and ZigBee. These are suitable for wearables because of their low cost, small-size transceivers and low power consumption. The recently developed IEEE 802.15.4a standard based on ultra-wideband (UWB) impulse radio opens the door for high-data-rate yet low-power, low-cost sensor network applications with the possibility of highly accurate location estimation.

For most monitoring applications, the sensor network would, in turn, have to transmit data to a remote site using an information gateway such as a mobile phone or laptop.